Reasoning test

For admission to Class 10

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 9 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 9.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, using ideas taught up to Class 9. Chapters are from the NCERT (CBSE) books a student joining Class 10 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Ganita Prakash

- *A Square and A Cube* – squares and cubes
- *Proportional Reasoning-1* – ratio and proportion
- *Fractions in Disguise* – percentages, profit and loss, discounts, interest
- *Proportional Reasoning-2* – ratio and proportion
- *Tales by Dots and Lines* – averages (mean)

Class 9 · Ganita Manjari

- *Measuring Space: Perimeter and Area* – perimeter and area
- *The Mathematics of Maybe: Introduction to Probability* – probability
- *Predicting What Comes Next?: Exploring Sequences and Progressions* – number patterns and series

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out,

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Curiosity

- *Exploring Forces* – floating, sinking and density; mass and weight
- *Pressure, Winds, Storms, and Cyclones* – force and pressure, pressure in liquids
- *Particulate Nature of Matter* – evaporation and condensation; change of state
- *Nature of Matter: Elements, Compounds, and Mixtures* – elements, compounds, mixtures and solutions
- *The Amazing World of Solutes, Solvents, and Solutions* – elements, compounds, mixtures and solutions
- *How Nature Works in Harmony* – food chains

Class 9 · Exploration

- *How Forces Affect Motion* – inertia, action and reaction
- *Work, Energy and Simple Machines* – energy changes
- *Exploring Mixtures and their Separation* – elements, compounds, mixtures and solutions
- *Sound Waves – Characteristics and Applications* – sound, and the speeds of light and sound

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Class 9 syllabus questions

Five maths and five science questions come from these Class 9 chapters only: the first-term chapters, because the test can be taken before the year ends.

Maths

Class 9 · Ganita Manjari

- *Ch 1. Orienting Yourself: The Use of Coordinates* – quadrants, distance, midpoint, section formula
- *Ch 2. Introduction to Linear Polynomials* – value and zero of a polynomial, linear growth
- *Ch 3. The World of Numbers* – irrational numbers, decimal expansions
- *Ch 4. Exploring Algebraic Identities* – identities, factorising
- *Ch 5. I'm Up and Down, and Round and Round* – circles: chords, angles, cyclic quadrilaterals

Science

Class 9 · Exploration

- *Ch 2. Cell – The Building Block of Life* – cell parts, plant and animal cells, osmosis
- *Ch 3. Tissues in Action* – plant and animal tissues
- *Ch 4. Describing Motion Around Us* – displacement, speed, acceleration, graphs, equations of motion
- *Ch 5. Exploring Mixtures and their Separation* – separating mixtures
- *Ch 6. How Forces Affect Motion* – Newton's laws, momentum

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   What is the angle between the hands of a clock at exactly 3 o'clock?
   A 45° **B** 60°
   C 75° **D** 90°

2. ENGLISH · BASICS
   The general's plan was simple but ingenious. As used here, "ingenious" most nearly means
   A dangerous **B** clever
   C costly **D** careless

3. SCIENCE · BASICS
   As a ball falls freely towards the ground, which energy change takes place?
   A potential energy to kinetic energy **B** kinetic energy to potential energy
   C kinetic energy to chemical energy **D** heat energy to potential energy

4. MATHS · CLASS 9
   On which axis does the point (0, −4) lie?
   A the x-axis **B** the y-axis
   C neither axis **D** both axes

5. ENGLISH · BASICS
   Choose the correct words: The bundle of letters ___ on the table.
   A were lying **B** are lying
   C was lying **D** have lain

6. SCIENCE · CLASS 9
   A train covers 150 km in 2.5 hours. What is its average speed, in km/h?
   Type-in answer: _____

7. MATHS · BASICS
   The price of a bag rises from ₹250 to ₹300. What is the percentage increase?
   A 15% **B** 20%
   C 25% **D** 50%

8. ENGLISH · BASICS
   Ear is to hearing as eye is to ___
   A tears **B** face
   C colour **D** sight

9. SCIENCE · BASICS
   An astronaut has a mass of 60 kg on Earth. What is her mass on the Moon?
   A 0 kg **B** 10 kg
   C 60 kg **D** 360 kg

10. MATHS · CLASS 9
    If p(x) = 4x + 7, what is the value of p(3)?
    Type-in answer: _____

11. ENGLISH · BASICS
After the heavy rain, the hill path became treacherous. As used here, "treacherous" most nearly means

A dangerous
B narrow
C muddy
D steep

12. SCIENCE · CLASS 9
Which part of a plant cell contains chlorophyll and makes food?

A Nucleus
B Chloroplast
C Vacuole
D Cell wall

13. MATHS · BASICS
A car uses 8 litres of petrol to travel 96 km. How many litres will it need to travel 156 km?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: The train arrived ___ the station on time.

A in
B on
C at
D to

15. SCIENCE · BASICS
An ant bite injects an acid into the skin. Which of these would best relieve the sting?

A a few drops of vinegar
B baking soda paste
C fresh lemon juice
D plain salt water

16. MATHS · CLASS 9
What do you get when you expand $(a - 3)(a + 3)$?

A $a^2 - 9$
B $a^2 + 9$
C $a^2 - 6a - 9$
D $a^2 - 3$

17. ENGLISH · BASICS
Author is to book as composer is to ___

A orchestra
B piano
C stage
D music

18. SCIENCE · CLASS 9
Which method recovers the salt from salt water?

A Filtration
B Decantation
C Evaporation
D Chromatography

19. MATHS · BASICS
₹5,000 is deposited at 10% interest per year, compounded yearly. What will the amount be after 2 years?

A ₹5,500
B ₹6,000
C ₹6,050
D ₹6,500

20. ENGLISH · BASICS
"The bus was late, the rain soaked us and the shop was shut. What a perfect day!" The speaker's tone is

A sarcastic
B sincere
C hopeful
D indifferent

21. SCIENCE · BASICS
40 g of sugar is dissolved in water to make 500 ml of solution. What is the concentration, in grams per litre?

Type-in answer: _____

22. MATHS · CLASS 9

How far is the point (6, 8) from the origin?

Type-in answer: _____

23. ENGLISH · BASICS

What comes next in the series? B, E, H, K, ___

A M
B N
C O
D L

24. SCIENCE · CLASS 9

A net force of 12 N acts on a 4 kg trolley. What is its acceleration?

A 0.33 m/s^2
B 3 m/s^2
C 16 m/s^2
D 48 m/s^2

25. MATHS · BASICS

In a class of 50 students, 30 play cricket, 25 play football and 8 play neither game. How many play both?

A 8
B 13
C 17
D 22

26. ENGLISH · BASICS

"New traffic lights were installed in May and road accidents fell in June, so the lights prevent accidents." Which fact, if true, most weakens this argument?

A The lights cost the town a lot
B Some drivers dislike waiting at lights
C Schools closed in June, cutting traffic
D Accidents also happen late at night

27. SCIENCE · BASICS

A burn from steam at 100 °C is usually worse than one from boiling water at 100 °C. This is because steam

A is at a higher temperature than water
B moves faster and hits the skin harder
C carries germs that the water does not
D gives out extra heat as it condenses

28. MATHS · CLASS 9

AB is a diameter of a circle and C is a point on the circle. If ∠CAB = 35°, what is ∠CBA?

A 35°
B 45°
C 55°
D 90°

29. ENGLISH · BASICS

No soldier is careless. Some cadets are careless. Which statement must be true?

A No cadet is a soldier
B Some cadets are not soldiers
C All soldiers are cadets
D Some soldiers are careless

30. SCIENCE · CLASS 9

A cyclist moving at 4 m/s speeds up uniformly at 0.5 m/s^2 for 6 seconds. What is the final velocity, in m/s?

Type-in answer: _____

Answers

1	**D. 90°**	The clock face is 360°, and each hour mark is 30° apart. From 12 to 3 is three hour marks, so the angle is 3 × 30° = 90°.
2	**B. clever**	Ingenious means clever and inventive. "Simple but" signals a quality that is praised, which rules out careless and dangerous.
3	**A. potential energy to kinetic energy**	As the ball falls it loses height, so it loses potential energy, and it speeds up, so it gains kinetic energy.
4	**B. the y-axis**	Its x-coordinate is 0, so the point is on the y-axis, 4 units below the origin. Points on the x-axis have y-coordinate 0 instead.
5	**C. was lying**	The subject is "the bundle", which is singular; "of letters" only describes it. So the verb is "was lying".
6	**60**	Average speed is total distance divided by total time: 150 km ÷ 2.5 h = 60 km/h. This is a type-in question: the answer is typed as a number.
7	**B. 20%**	The rise is ₹50, and the percentage is worked out on the original price: 50 ÷ 250 = 0.2, which is 20%.
8	**D. sight**	The pair links an organ to the sense it provides. The ear gives hearing and the eye gives sight.
9	**C. 60 kg**	Mass is the amount of matter and does not change from place to place. Her weight on the Moon is about one sixth, but her mass stays 60 kg.
10	**19**	Put x = 3 into the polynomial: 4 × 3 + 7 = 12 + 7 = 19. This is a type-in question: the answer is typed as a number.
11	**A. dangerous**	Treacherous means full of hidden danger. The rain may well have made the path muddy, but that is a cause, not the meaning of the word.
12	**B. Chloroplast**	Chloroplasts hold the green pigment chlorophyll and carry out photosynthesis. The nucleus controls the cell, the vacuole stores, the cell wall supports.
13	**13**	The car runs 96 ÷ 8 = 12 km on each litre, so 156 km needs 156 ÷ 12 = 13 litres. This is a type-in question.
14	**C. at**	We arrive "at" a place such as a station, and "in" a city or country. "Arrived to" is a common slip, but it is not standard English.
15	**B. baking soda paste**	An acid is neutralised by a base. Baking soda is a mild base; vinegar and lemon juice are acids and would not help.
16	**A. $a^2 - 9$**	The identity $(x - y)(x + y) = x^2 - y^2$ gives $a^2 - 3^2 = a^2 - 9$. The middle terms $-3a$ and $+3a$ cancel, so there is no a-term.
17	**D. music**	An author creates a book; a composer creates music. An orchestra and a piano perform it, and a stage is where it is performed.
18	**C. Evaporation**	Salt is dissolved, so it passes through filter paper and cannot be poured off. Evaporating the water leaves the salt behind.

19	**C. ₹6,050**	After year 1: 5,000 + 500 = ₹5,500. Year 2 earns 10% of 5,500, which is ₹550, giving ₹6,050. ₹6,000 is simple interest.
20	**A. sarcastic**	Everything described went wrong, so calling it a perfect day means the opposite of what is said. That is sarcasm.
21	**80**	500 ml is half a litre. 40 g in half a litre is the same as 80 g in a full litre. This is a type-in question.
22	**10**	The distance from the origin is $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. This is a type-in question: the answer is typed as a number.
23	**B. N**	Each letter is three places after the one before: B (+3) E (+3) H (+3) K (+3) N. M and L come from counting two or one place instead of three.
24	**B. 3 m/s²**	From F = ma, a = F ÷ m = 12 N ÷ 4 kg = 3 m/s². 48 multiplies instead of dividing, and 0.33 divides the wrong way round.
25	**B. 13**	50 − 8 = 42 students play at least one game. 30 + 25 = 55 counts the players of both games twice, so 55 − 42 = 13 play both.
26	**C. Schools closed in June, cutting traffic**	If traffic itself fell in June, that alone could explain fewer accidents, so the lights may not be the cause. The others do not touch the cause.
27	**D. gives out extra heat as it condenses**	Both are at 100 °C, but steam releases a large amount of extra heat (latent heat) when it turns back into water on the skin.
28	**C. 55°**	The angle in a semicircle is 90°, so ∠ACB = 90°. The angles of triangle ABC add to 180°, so ∠CBA = 180° − 90° − 35° = 55°.
29	**B. Some cadets are not soldiers**	The careless cadets cannot be soldiers, so at least some cadets are not soldiers. Nothing rules out other cadets being soldiers, so "no cadet is a soldier" goes too far.
30	**7**	Using v = u + at: v = 4 + 0.5 × 6 = 4 + 3 = 7 m/s. This is a type-in question: the answer is typed as a number.